

VISHAL YADAV

Mumbai, Maharashtra, India | vishalyadav.analytics@gmail.com | +91 9619039381
portfolio-one-sigma-39.vercel.app | github.com/Vishal123-tech

PROFESSIONAL SUMMARY

Data Science and Analytics fresher with hands-on experience in end-to-end ML pipelines, predictive modeling, EDA, ETL development, and business intelligence reporting. Proficient in Python, SQL, Power BI, and Tableau. Built and tuned classification and regression models achieving **90%+ accuracy**. Passionate about transforming large datasets into actionable business insights through data-driven solutions and compelling visualizations.

TECHNICAL SKILLS

Languages	Python, SQL, R (Foundational)
ML & Algorithms	Classification, Regression, Clustering (K-Means), Feature Engineering, Hyperparameter Tuning, Cross-Validation, SHAP Interpretability, EDA
ML Libraries	Scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy, Optuna
Visualization & BI	Power BI, Tableau, Matplotlib, Seaborn
Databases	PostgreSQL, MySQL
ETL & Pipelines	Data preprocessing, feature transformation, pipeline automation, model serialization
Tools & Platforms	Excel (Pivot Tables, Power Query, XLOOKUP), Jupyter Notebook, Git, GitHub

PROFESSIONAL EXPERIENCE

Deloitte	Jul 2025
<i>Data Analytics Virtual Intern — Forage</i>	<i>Remote</i>
- Built structured ETL processes in Excel and Tableau to clean, transform, and validate raw business datasets, ensuring data integrity for downstream analytics. - Analyzed large-scale client operational data to surface time-series trends and anomalies, producing data-driven recommendations that informed business process improvements. - Designed interactive Tableau dashboards communicating KPIs and performance narratives to stakeholder audiences, improving reporting clarity and decision-making speed. - Collaborated cross-functionally to align analytical outputs with business requirements and maintain consistent data quality standards across multiple datasets.	

PROJECTS

Diabetes Classification Python, Scikit-learn, XGBoost, LightGBM, Optuna	2026
- Built a full end-to-end ML pipeline on the Pima Indians Diabetes dataset (768 records) covering data ingestion, feature engineering, model training, and evaluation. - Engineered BMI categories and age bins; applied statistical outlier detection on Insulin and SkinThickness, improving baseline accuracy by ~14 percentage points. - Benchmarked 7 algorithms (LR, KNN, CART, RF, GBM, XGBoost, LightGBM); LightGBM + Optuna tuning achieved the top accuracy of 90.07%. - Applied SHAP analysis to identify Glucose, Insulin, and BMI as primary predictors; delivered visualizations of distributions, outliers, and class balance.	
Customer Segmentation & Value Analysis Python, Pandas, K-Means, Power BI, Seaborn	2025
- Designed an unsupervised analytics pipeline using K-Means clustering on transactional data to identify customer segments by spending frequency and monetary value. - Engineered RFM-style features (Recency, Frequency, Monetary) with Pandas to enhance cluster separability and improve business interpretability of segments. - Isolated high-value cohorts driving the majority of revenue; produced Power BI dashboards to support targeted marketing and customer retention strategies.	

AI Impact on Jobs & Layoff Risk Analysis | *Python, Pandas, Seaborn, Matplotlib, EDA*

2026

- Curated and analyzed a dataset of 9,000+ job records across 15+ industries to quantify AI automation exposure scores and layoff risk levels by role, sector, and geography.
- Performed in-depth EDA — uncovering that roles in IT, Finance, and Manufacturing face the highest displacement risk — and visualized trends using Seaborn and Matplotlib for stakeholder-ready reporting.
- Engineered derived features including risk tier classification and industry vulnerability index from raw job market data, enabling multi-dimensional analysis of AI's economic impact.
- Published a structured, reproducible dataset and end-to-end analysis notebook on GitHub/Kaggle, contributing an open resource for workforce strategy and AI policy research.

CERTIFICATIONS

Data Analytics Job Simulation — Deloitte via Forage	Jul 2025
Machine Learning for Data Science Projects — IBM SkillsBuild	Mar 2026
Data Classification — IBM SkillsBuild	Mar 2026
Artificial Intelligence Fundamentals — IBM SkillsBuild	Mar 2026
Google Analytics Certification — Google Skillshop	Jan 2026

KEY ACHIEVEMENTS

- Achieved **90.07% accuracy** on the Diabetes Classification pipeline by engineering features and tuning LightGBM with Optuna, a ~14 pp improvement over the baseline.
- Delivered **lowest RMSE and highest R^2** among 7+ regression models in the Insurance Cost project through systematic cross-validated hyperparameter search.
- Completed **4 end-to-end data science projects** across supervised learning, clustering, SQL analytics, and SHAP interpretability, all published on GitHub.
- Earned **4 industry certifications** in ML, AI, and Data Analytics from IBM SkillsBuild and Deloitte, demonstrating consistent upskilling commitment.

EDUCATION

Shree L R Tiwari College of Engineering

B.Tech in Electronics and Telecommunication Engineering

2023 – 2027

Mumbai, Maharashtra